AgriEYE 360: cotton field pipeline

AgriEYE 360 turns 360° photos of a cotton plot into results for each plant:

- open bolls and closed bolls
- leaf health (healthy or diseased leaves, % diseased)
- growth from one visit to the next
- a Street View style tour of the field

Last updated: 9 Nov 2026

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Contents

1. Status and results 3

1a. Field results (growth.csv) . . . 3

1b. Disease classifier v2: performance . . . 4

1c. Boll classifier (Roboflow) and the cloud problem . . . 4

1d. Which models are used (final choice) . . . 5

1e. SAM 3 leaf outlines: hand check . . . 5

1f. One-step boll detector (Roboflow): comparison . . . 5

1g. Roboflow datasets (all built from the 4-POV views) . . . 6

1h. Plant ID fixes (6 Nov) . . . 6

1i. Lessons learned . . . 7

2. Commands (Windows cmd) 7

2a. Field pipeline (agrieye.py) . . . 7

2b. Bolls: sorting, Roboflow datasets, Roboflow model . . . 8

2c. 4-POV training datasets (all models) . . . 8

2d. Leaves and disease (pipeline.py and helpers) . . . 8

2e. This guide . . . 9

Still open . . . 9

3. Architecture 9

Data flow . . . 9

Scripts (all in F:\Project and completed courses\Agri EYE 360\) . . . 10

Input data . . . 11

F:\Project and completed courses\Agri EYE 360\agrieye_output\ (field results) . . . 11

F:\Project and completed courses\Agri EYE 360\disease_autolabel\ (leaf and disease work) . . . 12

Other result folders . . . 12

4. Environment 12

5. Key charts 13

Disease classifier v2: results per epoch . . . 13

Disease classifier v2: dashboard . . . 13
Plant (station) positions after the 6 Nov fix . . . 14
Open bolls per plant over time (hand labels + Roboflow boll model) . . . 14
Closed bolls per plant over time . . . 15
% diseased leaves per plant over time . . . 15
Hand labels moved from the panorama onto the 4-POV views (check sheet) . . . 16
Roboflow boll model: every mistake on the 355 test images . . . 16
Open (cyan, solid) and closed (magenta, dashed) bolls on the 4 views with the most open bolls . . . 17
SAM 3 leaf outlines (P09, 31 Jul, east view): every outline is a real leaf, but some leaves have none . . . 17
One-step boll detector on a test view (cyan = open, magenta = closed, white = hand labels) . . . 18
R1 boll classifier: confusion matrix on 355 test crops . . . 18
R2 boll detector: confusion matrix on 18 test views . . . 19
SAM 3 alone, disease-spot prompts: highest-scored leaves (top) and diseased leaves it missed (bottom) . . . 19
L2 disease classifier: training per epoch . . . 20
L1 leaf finder: training per epoch . . . 20
Local boll detector (YOLO11n, 30 epochs, 1280 px): training per epoch . . . 21
Boll detector: 20 vs 25 vs 30 epochs (validation curves and test scores) . . . 21

1. Status and results

Stage	Status	Result
Import 360 photos	● Done	35 unique panoramas from 6 data folders (765 files; the rest were copies). 5 visits: 22 Jul (1 photo, no plant ID), 31 Jul, 4 Aug, 6 Aug, 20 Aug 2026
Plant IDs (stations)	● Done, checked 6 Nov	14 plants (P01–P13). Repeat shots removed from the analysis. At most 1 photo per plant per visit (30 photos used)
Slice into 4 views (N/E/S/W)	● Done	1440 px, 90° views, north-aligned. Analysis: 120 views. Training data: 140 views (all 35 photos)
Leaf crops (SAM 3) + manual sorting	● Done	diseased 1,259 · healthy 5,424 · rejected 7,231
Disease classifier v2 (YOLO11n-cls)	● Trained 6 Nov	89.2% accuracy on 835 field test leaves, ROC AUC 0.944. Only 22% on outside close-up photos (does not generalise)
Leaf finder (YOLO11n-seg)	● Trained 6 Nov	Mask precision 0.69, recall 0.62, mAP50 0.67, mAP50-95 0.40
SAM 3 leaf segmentation (text prompt "leaf")	● Done	5,085 leaves in 120 views, median score 0.57
SAM 3 leaf outlines, hand check	● Done 9 Nov	426 outlines on 10 views: **all real leaves** (precision 100%), 423 with a good fit (99.3%). Recall not measured (see 1e)
SAM 3 + YOLO v2 disease, whole field	● Done	4,231 leaves judged. % diseased: 31 Jul 15.3 · 4 Aug 21.4 · 6 Aug 13.2 · 20 Aug 25.6
Boll labels: SAM 3 + hand sorting	● Done 8 Nov	3,068 boll boxes; 2,970 hand-checked: open 441 · closed 2,465 · not a boll 64
Boll classifier (Roboflow ViT-Base)	● Trained 8 Nov	**97.7% on 355 test images** (open recall 91.7%, closed 100%, sky/cloud rejected 51/51)
Boll detector (Roboflow, one step), comparison only	● Trained 9 Nov	**mAP@50 43.6%** on 18 test views (open AP 26.8%, closed AP 60.4%). Not used in the analysis (see 1f)
4-POV training datasets for Roboflow	● Done 8 Nov	5 datasets in `roboflow_export\` (see 1g)
Field analysis, report, Street View	● Done 8 Nov	Re-run with new plant IDs, leaf finder, disease v2 and boll labels (hand > Roboflow model > SAM 3)
Disease classifier v3 (field + public data)	● Data ready, not trained	`train_disease_v3.py train` (only when you decide to train)

1a. Field results (growth.csv)

Open bolls per plant (sum of the 4 views):

Plant	31 Jul	4 Aug	6 Aug	20 Aug
P02	0	3	13	27
P03	0	11	–	53
P06	0	6	–	53
P09	2	2	5	41
P10	–	3	1	66
P13	–	–	15	74

- **Bolls are opening, not disappearing:** closed bolls fall at the same time (P06 110 → 18, P02 71 → 24, P09 105 → 26).
- **Leaf disease rises by 20 Aug:** P09 34%, P13 31%, P03 29%, P06 26% diseased leaves (31 Jul: about 7–25%).

- **Boll labels used:** 2,557 from your hand sorting, 85 from the Roboflow model; 9 uncertain model answers (`check_bolls`).
- **Caveat:** the 4 Aug photos were taken in evening light, which makes red/orange leaves look diseased; compare 31 Jul, 6 Aug and 20 Aug first.

1b. Disease classifier v2: performance

Tested on 835 leaves from photos the model never saw during training. Every visit is included in the test set.

	Precision	Recall
Diseased	85.7%	82.7%
Healthy	91.0%	92.7%

- Confusion matrix: 239 diseased found, 50 missed, 40 healthy wrongly flagged, 506 healthy correct.
- **Training:** 2,924 training leaves, batch 32, image size 224; 19 epochs run (17 min on CPU), **best epoch 9** kept as `best.pt` (early stopping, patience 10). Leaf finder L1: 60 epochs (6 h), best epoch 59. Graphs: `sample_results raining_graphs\`.
- Accuracy by visit: 31 Jul 92% · 4 Aug (evening light) 85% · 6 Aug 94% · 20 Aug 88%.
- **"Needs checking" band:** leaves with a diseased probability between 0.2 and 0.8 get a yellow outline and are counted in `check_leaves`. The model is confident on 74% of leaves, and 94% of those are correct.
- **Can SAM 3 judge disease by itself? No (two tests).**
  - *Test 1, whole-leaf prompts* (`sam3_disease_test.py`, 444 leaves on photo tiles): "diseased leaf" picks almost every leaf (precision 43%, AUC 0.45); best rule 64% accuracy, AUC 0.75.
  - *Test 2, disease-spot prompts* (`sam3_spot_test.py`, all 835 test leaves, each enlarged to ~320 px): SAM 3 segments "brown spot", "leaf spot", "hole in leaf", "yellow patch"; score = share of the leaf covered by spots. Best: all spot prompts summed, AUC 0.691; "brown spot" alone 73% accuracy but finds only 37% of diseased leaves (always saying "healthy" gives 65%). "Leaf spot", "hole" and "yellow patch" are near chance (AUC 0.52-0.57).
  - Why: SAM 3 finds distinct spots, but many diseased leaves are discoloured all over (evenly brown, dried, curled) with no separate spot. Its most confident picks are mostly right (9 of the top 12).
  - Versus v2: 89.2% accuracy, AUC 0.944. So SAM 3 is used to **find** leaves and the trained classifier **judges** them. Files: `sample_results\sam3_spot_test\` (`report.txt`, `examples.jpg`).
- **Outside photos:** v2 scores 22% on public close-up leaf photos. Public CC BY 4.0 datasets (Mendeley t9hgvk2h9p and SAR-CLD-2024 b3jy2p6k8w) are prepared for v3 (2,681 leaf crops).

1c. Boll classifier (Roboflow) and the cloud problem

- **Pipeline:** SAM 3 finds every boll (prompts "open cotton boll", "closed cotton boll", "flower"), the boll classifier decides open / closed / not a boll.
- **Why not SAM 3's own label:** of the boxes SAM 3 called "open", only about half were open; "flower" was almost never a flower (red closed bolls of this red-leaf variety).
- **Hand sorting:** `sort_bolls_grid.py` shows 30 crops per page with SAM 3's guess filled in; you only click the wrong ones. Your sorting changed SAM 3's answer on 339 boxes.
- **Cloud vs cotton:** an early Roboflow model mixed up white clouds and open cotton. Fix: 383 sky/cloud patches (smooth, away from any boll) added as `not_boll`, plus no brightness augmentation.
- **Model:** Roboflow ViT-Base (ImageNet-21k), 224×224, flip + ±15° rotation, 3×, early stopping (0.16 credits). Roboflow id `cotton-boll-open-or-close-3frah/3`.
- **Test (355 images never used in training, `eval_roboflow_boll.py`):**

Class	Precision	Recall	Test images
open_boll	95.7%	91.7%	48
closed_boll	97.7%	100%	252
not_boll	100%	92.7%	55

- The 8 mistakes are bolls just starting to crack, tiny crops and one cloud-filled crop; 3 had low confidence (0.43–0.56). Answers below 0.6 are counted as `check_bolls`.
- **Limitation:** only 4 test images were real non-sky "not a boll" objects (all 4 wrong); open bolls are rare (48 in test).
- On all 3,068 field crops the model agrees with your hand labels 95.4% (includes training crops, so this is a consistency check, not a test score).

1d. Which models are used (final choice)

Every task is done in **two steps**: SAM 3 finds the objects, then a trained classifier judges each one.

Task	Step 1: find	Step 2: judge	Score	Trained where
Cotton bolls	SAM 3 (ready-made, not trained)	Boll classifier, open / closed / not a boll	97.7%	Roboflow website
Leaves and disease	SAM 3 / leaf finder (YOLO11n-seg)	Disease classifier v2, healthy / diseased	89.2%	Local machine

- Roboflow models: boll classifier (used) and boll detector (comparison only, see 1f).
- Local models: leaf finder `disease_autolabel\models\leaf_seg\weights\best.pt` and disease v2 `disease_autolabel\models\disease_cls\weights\best.pt`. `disease_cls_v1_backup\` is the older v1 of the same model. Disease v3 is not trained.

1e. SAM 3 leaf outlines: hand check

`sam3_seg_check.py` showed 10 views (spread over all 4 visits) with every SAM 3 leaf outline (prompt "leaf", conf 0.3).

- **Precision: 426 of 426 outlines (100%) were judged to be real leaves.** 423 had a good fit (99.3%); 3 were partial (all in P09, 31 Jul, east view: small dead leaves on the soil). No boll, stem, sky or soil was outlined. A few outlines are weed or grass leaves, not cotton.
- **Two rounds:** a precision round (`answers.json`) and a recall round (`answers_recall.json`, filled outlines, missed leaves to be marked). The report uses the recall round.
- **Recall: not measured.** Leaves with no outline were not marked. A visual check shows SAM 3 **misses some clearly visible leaves**, so recall is below 100%.
- **What this means:** SAM 3 gives a reliable **sample** of leaves on each plant. That is fine for **% diseased** (a ratio). **Leaf counts are not used as a measure of plant size.**
- Files: `F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\` (`report.txt`, `per_view.csv`, `answers.json`).

1f. One-step boll detector (Roboflow): comparison

- **Data:** `1_cotton_boll_detection` (140 views, about 3,000 boll boxes: open 453, closed 2,544; split 94 / 28 / 18 views), project "Cotton boll labelling".
- **Roboflow id:** `cotton-bolls-field-view-trbz1/1`. Training graphs: mAP@50 about 35% by epoch 40, about 41% (validation) after about 320 epochs; losses flat after about 30 epochs.
- **Test (18 views never used in training, 308 hand-checked boxes, `eval_roboflow_detect.py`):** Roboflow's free plan shows no metrics, so they were computed from the hosted model's answers (IoU 0.5).

Class	Boxes	AP@50	AP@50:95	Precision (conf 0.5)	Recall (conf 0.5)
open_boll	50	26.8%	11.0%	56.2%	18.0%
closed_boll	258	60.4%	32.4%	75.5%	43.0%
All (mAP)	308	**43.6%**	**21.7%**	73.6%	39.0%

- **Confusion matrix (conf 0.5, IoU 0.5):** of 308 bolls, 122 got a box and 120 of those had the right class (only 2 open bolls called closed, none the other way); 186 were missed (39 of 50 open, 147 of 258 closed); 41 boxes were drawn where no boll was labelled. So R2 mostly **misses** bolls rather than mixing up open and closed. Pictures: `sample_results\confusion_matrices\` (`confusion_matrix_bolls.py`, no API key needed).

- **Local YOLO11n detector, 30 epochs, image size 1280** (`train_boll_detector.py`, same 140 views and split, 53 min on CPU, best epoch 26). Test views (18): **mAP@50 52.5% with the same scoring as R2** (our script, duplicate boxes merged at overlap 0.5; open AP 42.5%, closed AP 62.5%), against **43.6% for R2** scored the same way, so +9 points, mainly on open bolls. **With Ultralytics' default settings** (overlap 0.7, as in the training stats) the same model gives **47.9% on the test views and 46.2% on the validation views**; the gap is duplicate boxes on touching bolls. The bigger input keeps small bolls visible. Still far below the two-step R1 (97.7%), so the pipeline is unchanged; the model is stored in `agrieye_output\boll_detector\` where `analyze` does not use it.
- **20 / 25 / 30 epochs** (same settings, 18 test views, Ultralytics default / same scoring as R2): 20 epochs 45.8% / 49.9%, 25 epochs 45.2% / 51.8%, **30 epochs 47.9% / 52.5%**. On validation the three tied within 1 point; the tie-breaker (mAP50-95) kept the **30-epoch model**. More epochs help a little, mostly for open bolls. Chart: `sample_results\training_graphs\boll_detector_epochs_compare.png`; reports `report_20ep.txt`, `report_25ep.txt`, `report_30ep.txt`.
- **Tiling did not help:** cutting each view into 2×2 tiles gave mAP@50 28.8% (more false closed bolls), so whole views are better for this model.
- **What it misses:** mostly small, far-away bolls; it finds the large bolls well. Open bolls are found least (9 of 50 at conf 0.5), against 91.7% open recall for the two-step classifier.
- Files: `F:\Project and completed courses\Agri EYE 360\sample_results
oboflow_boll_detect_test\ (report.txt, preview_*.jpg, predictions_*.csv`).
- **Why it is lower:** most bolls are only 20–40 px in a 1440 px view (smaller still at 640 px input); closed bolls are green on green leaves; only 140 photos (28 for validation, so the curve is noisy); the labels come from SAM 3 boxes, so real bolls that SAM 3 missed count as false detections.
- **Conclusion:** with this little data, **two steps (SAM 3 + classifier, 97.7%) work much better than one detector**. The analysis uses the two-step method.

1g. Roboflow datasets (all built from the 4-POV views)

Folder: `F:\Project and completed courses\Agri EYE 360\roboflow_export\`

The zip copies were deleted on 9 Nov 2026 to save space (0.9 GB); `make_roboflow_datasets.py` and `make_pov_dataset.py export` rebuild them, and Roboflow also accepts a whole folder upload.

Folder	Roboflow project type	Contents
`1_cotton_boll_detection\` (upload copy: `UPLOAD_cotton_bolls_labelled\`)	Object Detection	140 views, 3,004 boll boxes (2,906 hand-checked); split 94 / 28 / 18 views
`2_cotton_boll_classification\`	Classification (single-label)	open 441, closed 2,465, not_boll 64 + 383 sky; split train 2,328 / valid 670 / test 355
`3_leaf_segmentation\`	Instance Segmentation	140 views, about 11,200 leaf outlines
`4_leaf_disease_field\`	Classification	Field leaves cut from the views: diseased 816, healthy 2,263
`5_leaf_disease_field_outside\`	Classification	Dataset 4 + 2,681 public leaves

- **Labels moved from panorama to views:** the hand-labelled leaf outlines were projected with the exact inverse of the slicer's maths (checked visually). 10,108 leaves landed on a view; 3,785 were above or below the 45° view limit.
- **Split without leakage:** grouped by plant + visit (+ view direction for the boll sets; the 4 views do not overlap). Repeat shots always stay with their twin. The boll sets are balanced so open bolls are also 70 / 20 / 10.
- **Roboflow tips learned:** choose "Use Existing Values" for the split (never the random slider); Roboflow needs a test split of at least 2 images; the hosted API needs the **private** API key (the `rf_…` publishable key gives 401).

1h. Plant ID fixes (6 Nov)

- **Repeat shots blanked**, keeping the sharper first shot: P04 6 Aug 12:35 · P07 4 Aug 17:26 · P06 4 Aug 17:19:47 · P10 6 Aug 12:44 (same image as 12:43).

- **P06 split in two.** P12 is the stop on the walk back: 31 Jul 08:05 and 4 Aug 17:13.
- **North-east end assigned by walking order:** P09 first stop, P10 second, P13 third.
- Photos cannot confirm plant identity across visits (feature matching scores about the same for same and different plants). The split relies on: two separate stops on one walk are two plants, and the walking order is the same every visit (confirmed). Treat P06/P12 and P09/P10/P13 as best guesses.

1i. Lessons learned

- **Label noise was the main problem** for disease: about 1/3 of the first "diseased" crops were clean leaves in evening light. A strict second review raised diseased recall from 54% to 83%.
- **Hold-key accident:** 662 crops got the same label from a held key; re-checked, and all sorters now ignore a held key.
- **Same photo, two timestamps:** `360 files data\2.jpg` and `6.jpg` are the same pictures as P02/P11 on 31 Jul with EXIF times 30 s apart. Duplicates are matched by image content and the copy with the plant ID is kept.
- **Training images should look like the images the model will see:** all training data now comes from the 4-POV views, not the stretched panoramas.
- **Clouds look like open cotton** at low resolution: add sky examples, avoid brightness augmentation, and tile detection images in Roboflow.
- **A hand check must also look for what is missing:** in the SAM 3 outline check every outline starts green, so "all green" only tells you the outlines are right, not that every leaf was found.
- **Small objects favour two steps:** a one-step boll detector reached 43.6% mAP@50 on test views (open bolls 26.8% AP), while SAM 3 + a crop classifier reached 97.7%.
- **Panoramas:** the TOZO AnyVue 360 compass heading is correct (checked against the sun: east at 08:00, west at 17:00); GPS is only accurate to about 2–3 m.

2. Commands (Windows cmd)

Always use the venv Python. Open **cmd** and set two shortcuts first:

```
set PY="C:\Users\bhara\Downloads\.venv\Scripts\python.exe"
set P=F:\Project and completed courses\Agri EYE 360
```

2a. Field pipeline (`agrieye.py`)

One click: double-click `F:\Project and completed courses\Agri EYE 360\run_pipeline.bat` (import → stations → slice → autolabel → analyze → report → streetview, skips finished steps, log in `agrieye_output\pipeline_log.txt`).

```
%PY% "%P%\agrieye.py" import
%PY% "%P%\agrieye.py" stations
%PY% "%P%\check_stations.py"
%PY% "%P%\agrieye.py" slice
%PY% "%P%\agrieye.py" autolabel
%PY% "%P%\agrieye.py" analyze --use-sam3-labels
%PY% "%P%\agrieye.py" report
%PY% "%P%\agrieye.py" streetview
```

Command	What it does
`import`	Reads GPS, time and heading from the photos and writes `catalog.csv`
`stations`	Groups photos into plant IDs (`stations.csv`); your edits are kept unless you add `--force`
`check_stations.py`	Picture sheet and map for checking the plant IDs
`slice`	Cuts each 360 photo into N/E/S/W views
`autolabel`	SAM 3 boll/flower boxes on all views
`analyze`	Bolls (hand label > Roboflow model > SAM 3), leaves, % diseased, canopy, growth tables and plots

Command	What it does
`report`	Annotated views and one summary sheet per plant
`streetview`	Tour JSON for the field viewer, one per visit

2b. Bolls: sorting, Roboflow datasets, Roboflow model

```
:: 1. SAM 3 boxes for the extra (training-only) views, then cut crops for sorting
%PY% "%P%\boll_extra_views.py"
%PY% "%P%\make_boll_crops.py"

:: 2. Hand sorting: grid of 30, click to fix, Enter = accept page, U = undo
%PY% "%P%\sort_bolls_grid.py"

:: 3. Roboflow datasets 1 + 2 (boll detection, boll classification with sky negatives)
%PY% "%P%\make_roboflow_datasets.py"

:: 4. Roboflow model: set your PRIVATE key in this cmd window first (never save it in a file)
set ROBOFLOW_API_KEY=your_private_key
%PY% "%P%\eval_roboflow_boll.py"
%PY% "%P%\boll_classify_roboflow.py"
```

- `eval_roboflow_boll.py` tests the model on the 355 test images (`--limit 5` for a quick check) → `sample_results\roboflow_boll_test\`.
- `boll_classify_roboflow.py` classifies every SAM 3 boll box (about 0.2 credits, resumable) → `agrieye_output\objects\boll_cls_roboflow.csv`, used by `analyze`.

2c. 4-POV training datasets (all models)

```
%PY% "%P%\make_pov_dataset.py" all
```

Finds all unique 360 photos in every Data set folder, slices them, moves the hand leaf labels onto the views and writes Roboflow datasets 3–5 (and calls `make_roboflow_datasets.py` for 1–2).

2d. Leaves and disease (`pipeline.py` and helpers)

```
:: Leaf crops from the panoramas with SAM 3, sorting, reviews
%PY% "%P%\sam3_autolabel.py"
%PY% "%P%\sort_crops.py"
%PY% "%P%\find_clean_diseased.py"
%PY% "%P%\sort_crops.py" --review diseased
%PY% "%P%\find_suspect_healthy.py" --n 500
%PY% "%P%\sort_crops.py" --review healthy

:: Disease classifier and leaf finder (training: only when you decide to train)
%PY% "%P%\build_disease_dataset.py" --max-healthy 2500
%PY% "%P%\pipeline.py" traincls
%PY% "%P%\pipeline.py" leafdata
%PY% "%P%\pipeline.py" trainleaf
%PY% "%P%\model_metrics.py"

:: SAM 3 leaf segmentation of the field, alone or + YOLO disease
%PY% "%P%\sam3_segment_field.py"
%PY% "%P%\sam3_segment_field.py" --classify

:: SAM 3 on any image (text prompt), disease-prompt experiment, model tests
%PY% "%P%\vlm_segment.py" "<image or folder>" --prompt "leaf"
%PY% "%P%\sam3_disease_test.py" report
%PY% "%P%\test_all_models.py"
%PY% "%P%\verify_leaves.py"

:: Disease v3 with public data (data built; train only when you decide)
%PY% "%P%\prepare_external.py"
%PY% "%P%\train_disease_v3.py" build
%PY% "%P%\train_disease_v3.py" train
%PY% "%P%\train_disease_v3.py" compare
```

Every training run keeps the best model and redraws the graphs (`train_guard.py`). `traincls`, `trainleaf` and `train_disease_v3.py train` train into `models\<name>_new\`, then test the NEW and the CURRENT model on the same validation data and keep the better one as `models\<name>\` (the folder all scripts read):

Model	Decided by	Tie-breaker	Full tie
Disease classifier (`disease_cls`, `disease_cls_v3`)	accuracy on the validation leaves	diseased recall	current model kept
Boll detector (`boll_det`, `train_boll_detector.py`)	box mAP50 on the 28 validation views	box mAP50-95	current model kept
Leaf finder (`leaf_seg`)	mask mAP50 on the validation tiles	mask mAP50-95	current model kept

- **Differences under 1 point count as a tie** (noise on these small validation sets); then the tie-breaker decides, again with a 1-point margin.
- The other model is never deleted: `models\<name>_previous_<date>\` (replaced) or `models\<name>_rejected_<date>\` (not better).
- Every decision is logged in `disease_autolabel\models raining_history.csv`; the copies in `Results\models\` are refreshed.
- `training_graphs.py` then redraws `sample_results raining_graphs\` (L1, L2 and v3 once trained) from the active models' logs.
- v3 never replaces v2 automatically: compare them with `train_disease_v3.py compare`.
- Re-run the decision by hand: `%PY% "%P% rain_guard.py" decide disease_cls`.

2e. This guide

```
%PY% "%P%\make_readme_pdf.py"
```

Still open

1. Optional: SAM 3 recall. Run `sam3_seg_check.py` again (after renaming `answers.json`) and right-click every clear leaf with no outline.
2. Optional: tiled test of the Roboflow boll detector on the 18 test views (open and closed scored separately).
3. Optional: train the leaf disease model on Roboflow (folder `roboflow_export\5_leaf_disease_field_outside\`) or v3 locally, and compare with v2.

SAM 3 hand check, after sorting:

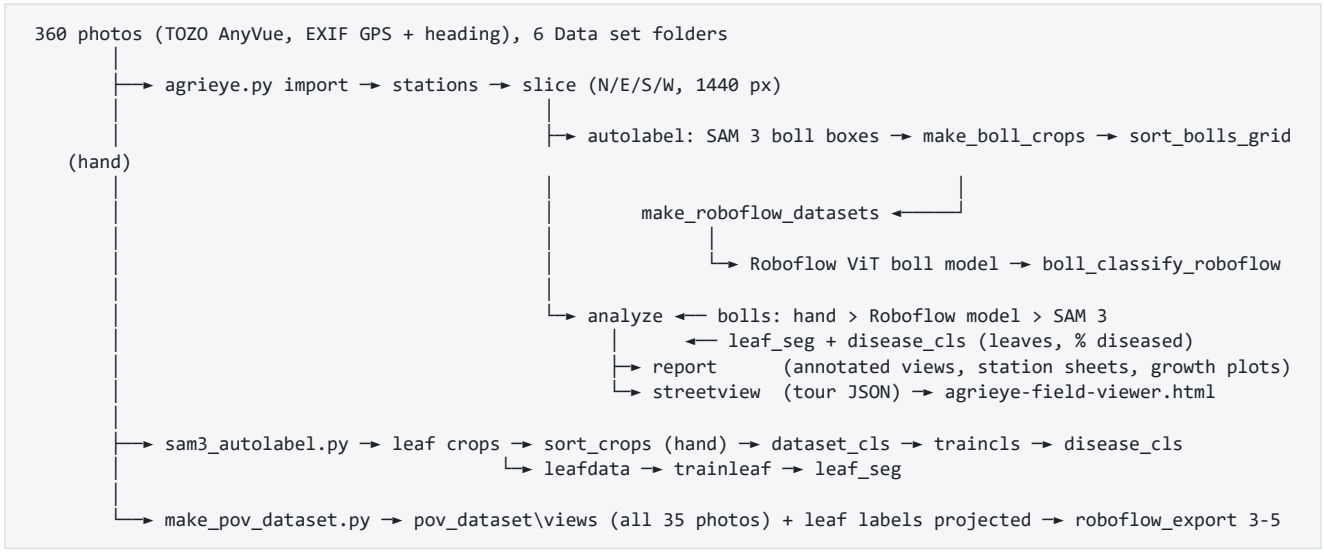
```
%PY% "%P%\sam3_seg_check.py"
%PY% "%P%\sam3_seg_check.py" --report
```

Clear open / closed boll pictures (cyan solid = open, magenta dashed = closed):

```
%PY% "%P%\make_boll_figures.py"
%PY% "%P%\make_boll_figures.py" --plant P13
```

3. Architecture

Data flow



Scripts (all in `F:\Project and completed courses\Agri EYE 360\`)

File	Purpose
`agrieye.py` + `agrieye_config.json`	Field pipeline: import, stations, slice, autolabel, refine, traindet, analyze, report, streetview, all
`run_pipeline.bat`	One-click field pipeline with log
`pipeline.py`	Leaf and disease pipeline: status, presort, leafdata, trainleaf, traincls, prelabel, predict
`sam3_autolabel.py`	SAM 3 leaf segmentation of the panoramas into crops and manifest
`sort_crops.py`, `find_clean_diseased.py`, `find_suspect_healthy.py`, `score_uncertain.py`	Leaf sorting and review tools
`build_disease_dataset.py`, `model_metrics.py`	Disease dataset and performance charts
`check_stations.py`	Plant-ID picture sheet, map and warnings
`make_boll_crops.py`, `sort_bolls.py`, `sort_bolls_grid.py`, `check_flowers.py`	Boll crops, sorters (one-by-one and grid), flower check sheets
`boll_extra_views.py`	SAM 3 boll boxes for the training-only views (22 Jul photo)
`make_pov_dataset.py`	All 35 photos → 4-POV views, leaf labels projected, Roboflow datasets 3–5
`make_roboflow_datasets.py`, `export_roboflow_bolls.py`	Roboflow datasets 1–2, 70/20/10 split, sky negatives
`eval_roboflow_boll.py`	Tests the Roboflow boll model on the test split
`eval_roboflow_detect.py`	Tests the Roboflow boll detector (R2) on the 18 test views: mAP, precision, recall per class
`list_roboflow_models.py`	Lists every project and version in the Roboflow workspace
`confusion_matrix_bolls.py`	Confusion matrices of R1 and R2 from saved test answers (no key, no credits)
`training_graphs.py`	Roboflow-style training graphs (performance + losses per epoch) for L1 and L2 from their training logs
`train_guard.py`	After every training run: keep the better of new vs current model, log it, redraw the graphs
`train_boll_detector.py`	Local YOLO11n boll detector (train / test on the 18 test views, same scoring as R2)
`boll_classify_roboflow.py`	Classifies every boll box with the Roboflow model for `analyze`
`sam3_segment_field.py`	SAM 3 leaf segmentation of the field (`--classify` adds YOLO disease)

File	Purpose
vlm_segment.py	SAM 3 text-prompt segmentation of any image
sam3_disease_test.py	Experiment: can SAM 3 judge disease by itself (no)
sam3_spot_test.py	Experiment 2: SAM 3 segments disease spots on enlarged leaves (AUC 0.69, no)
sam3_seg_check.py	Hand check of SAM 3 leaf outlines (good / partial / wrong, missed leaves) → precision and recall
make_boll_figures.py	Open / closed boll pictures for the thesis (cyan solid / magenta dashed)
prepare_external.py, train_disease_v3.py	Public data and disease v3 (field + outside)
test_all_models.py, verify_leaves.py, review_leaf_sorting.py	Model tests and label checks
make_readme_pdf.py	Builds AgriEYE360_project_guide.pdf from this README
sam3.pt	SAM 3 weights (3.4 GB; do not put in GitHub)
AgriEYE st.Viwer\agrieye-field-viewer.html	Street View viewer that loads the tour JSON

Input data

Location	Contents
F:\Project and completed courses\Agri EYE 360\AgriEye 360 field data\Data set\overall dataset\	The 34 panoramas used for the analysis
...\Data set\360 files data\, fourth iteration\, Second 100 data sets\, Third new data set\	Other copies; add the 22 Jul photo for training
F:\Project and completed courses\Agri EYE 360\external_data\	Public CC BY 4.0 leaf datasets (downloads, extracted, crops)

F:\Project and completed courses\Agri EYE 360\agrieye_output\ (field results)

Location	Contents
...\agrieye_output\catalog.csv, stations.csv, stations_backup_auto.csv	Photos, **plant ID for each photo** (edit station), automatic IDs before the fix
...\agrieye_output\station_check\	Plant-ID sheet and GPS map
...\agrieye_output\povs\P01\ … P13\, _skipped\	4 views per photo used in the analysis; views of blanked repeat shots
...\agrieye_output\objects\sam3_labels.jsonl, sam3_labels_extra.jsonl	SAM 3 boll boxes (analysis views; training-only views)
...\agrieye_output\objects\boll_cls_roboflow.csv, boll_cls_vs_hand.txt	Roboflow model answer for every boll box; agreement with hand labels
...\agrieye_output\boll_crops\open\, closed\, not_boll\, unsorted\, manifest.csv	Hand-sorted boll crops
...\agrieye_output\pov_dataset\	Training views of all 35 photos, catalog.csv, split.csv, leaves.jsonl, check_projection.jpg
...\agrieye_output\sam3_segmentation\	SAM 3 leaf outlines of the field
...\agrieye_output\sam3_yolo_disease\	SAM 3 leaves judged by disease v2 (summary.csv)
...\agrieye_output\results\detections.csv, growth.csv	**Results per view and per plant + visit**
...\agrieye_output\results\plots\	Growth charts (open/closed bolls, leaves, % diseased, canopy, check columns)
...\agrieye_output\results\povs\, station_sheets\, leaf_health\	Annotated views, one sheet per plant, leaf-health pictures
...\agrieye_output\streetview\tour_2026-07-31.json … tour_2026-08-20.json	Street View tours, one per visit

`F:\Project and completed courses\Agri EYE 360\disease_autolabel\` **(leaf and disease work)**

Location	Contents
`...\disease_autolabel\manifest.jsonl`	One line per leaf crop: source photo, box, outline, score
`...\disease_autolabel\crops\diseased\`, `healthy\`, `rejected\`	Hand-sorted leaf crops (1,259 / 5,424 / 7,231)
`...\disease_autolabel\dataset_cls\`, `dataset_cls_v3\`, `dataset_leaf\`, `dataset_seg\`	Classifier v2 data, v3 data (field + outside), leaf finder tiles, healthy/diseased segmentation
`...\disease_autolabel\models\disease_cls\weights\best.pt`	**Disease classifier v2 (current)**; v1 backup in `disease_cls_v1_backup\`
`...\disease_autolabel\models\leaf_seg\weights\best.pt`	**Leaf finder**
`...\disease_autolabel\metrics\`	Classifier charts (`00_results_grid.png`, `dashboard.png`, …)

Other result folders

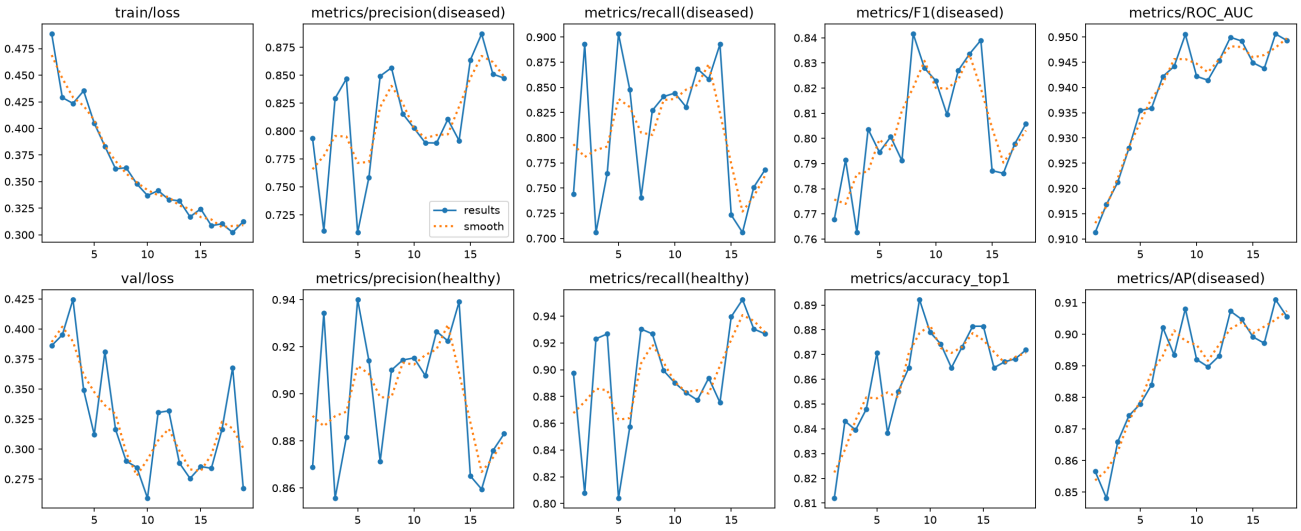
Location	Contents
`F:\Project and completed courses\Agri EYE 360\roboflow_export\`	The 5 Roboflow datasets (folders + zips), `2_sky_negatives_preview.jpg`
`F:\Project and completed courses\Agri EYE 360\sample_results\roboflow_boll_test\`	Roboflow boll model test: `report.txt`, `mistakes.jpg`, `predictions.csv`
`F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\`	SAM 3 leaf outline check: `report.txt`, `per_view.csv`, `answers.json`
`F:\Project and completed courses\Agri EYE 360\sample_results\boll_figures\`	Open / closed boll pictures (`00_open_vs_closed.jpg`, one file per view)
`F:\Project and completed courses\Agri EYE 360\sample_results\`	Sample outputs, model tests, SAM 3 experiments
`F:\Project and completed courses\Agri EYE 360\Results\`	Workspace links to the segmentation folders and models (junctions: delete with `rmdir`, never `Remove-Item -Recurse`)

4. Environment

- **PC:** Windows 11, Intel i5-1155G7, 16 GB RAM, CPU only (no GPU).
- **Python:** `C:\Users\bhara\Downloads\.venv\Scripts\python.exe` (ultralytics with SAM 3, OpenCV, matplotlib, inference-sdk for Roboflow).
- **Roboflow:** workspace `bharaths-workspace-3fhlw`, free plan (public projects, about 10 credits a month). Keep the private API key out of files, screenshots and GitHub.
- **Before long runs:** set sleep to "never" so the PC stays awake.

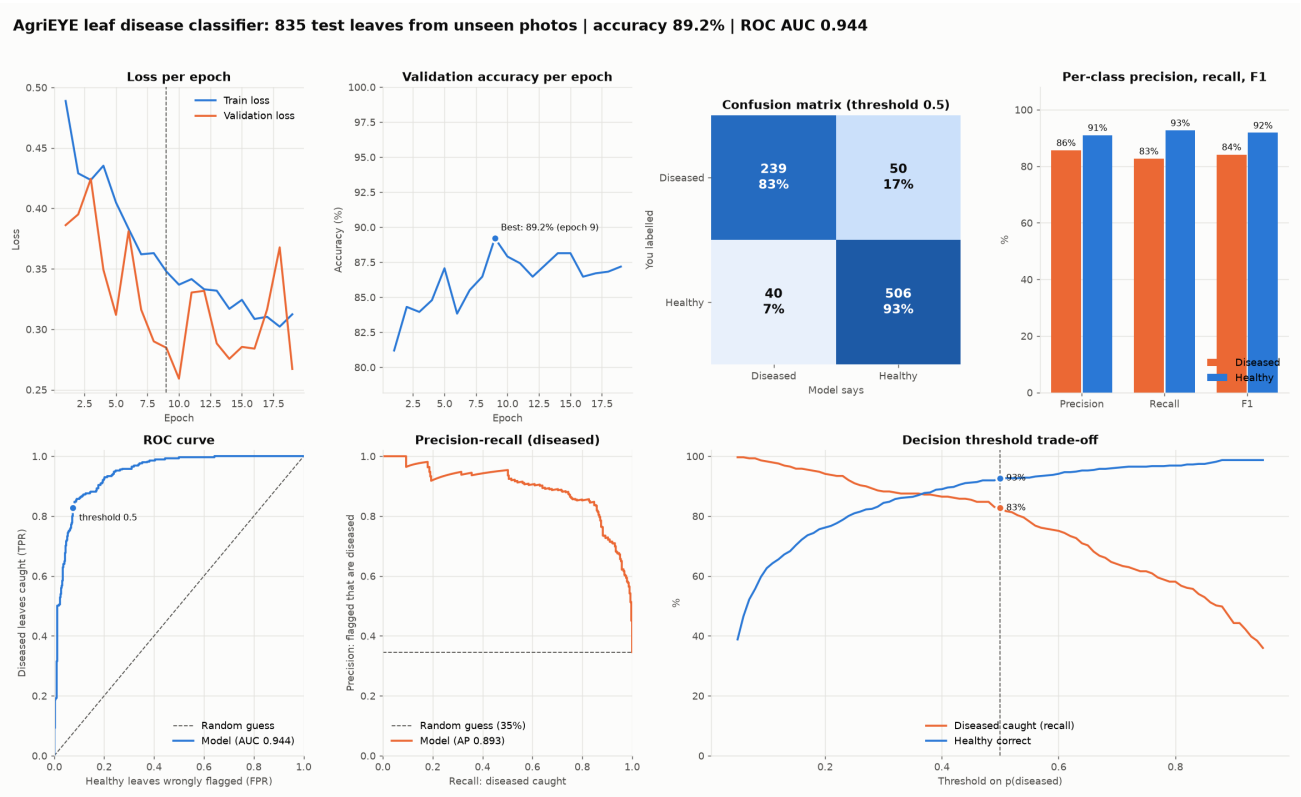
5. Key charts

Disease classifier v2: results per epoch



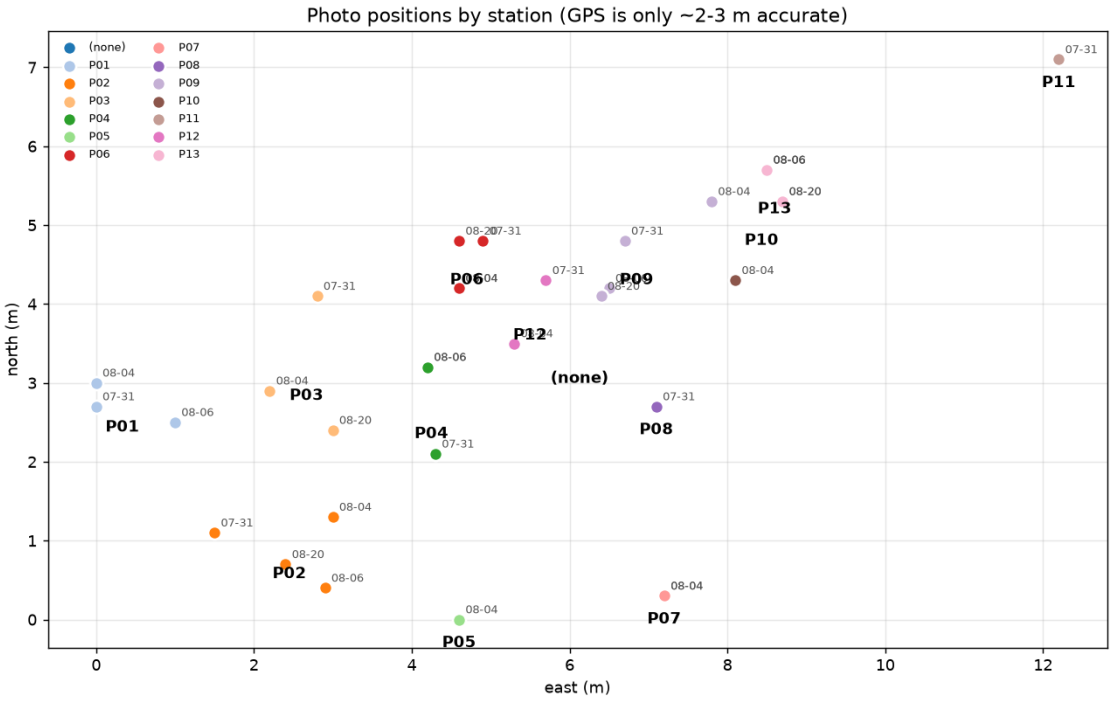
File: F:\Project and completed courses\Agri EYE 360\disease_autolabel\metrics\00_results_grid.png

Disease classifier v2: dashboard



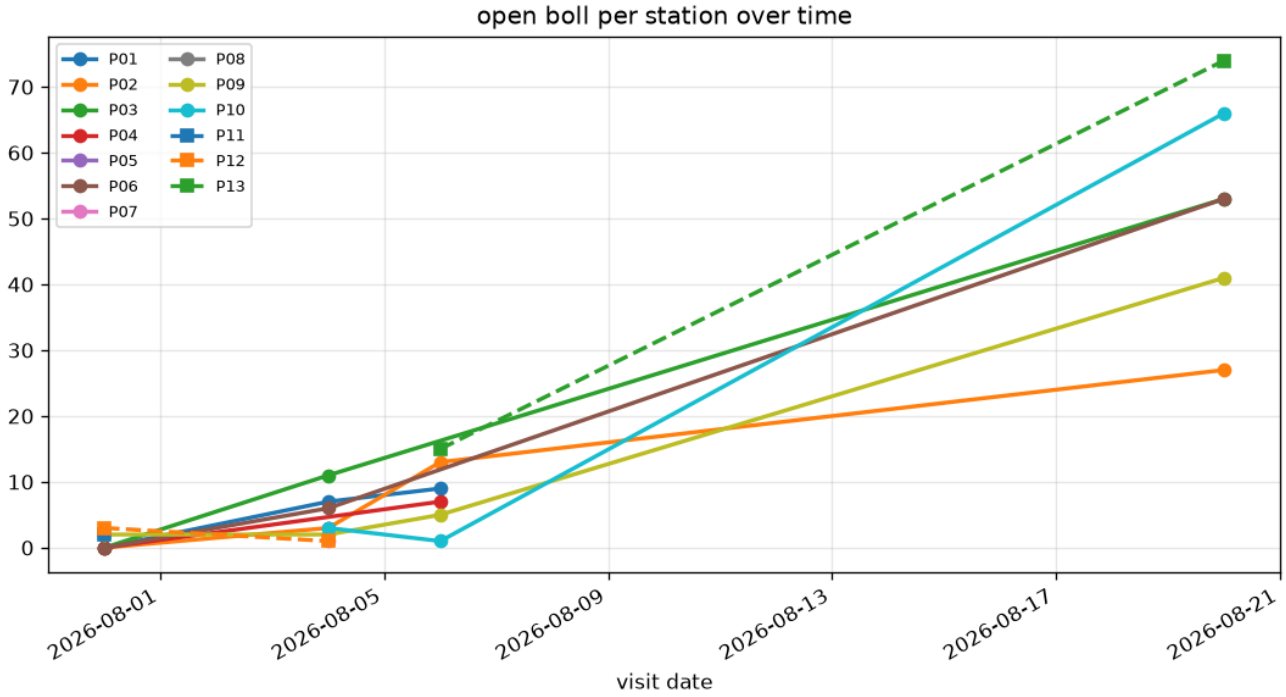
File: F:\Project and completed courses\Agri EYE 360\disease_autolabel\metrics\dashboard.png

Plant (station) positions after the 6 Nov fix



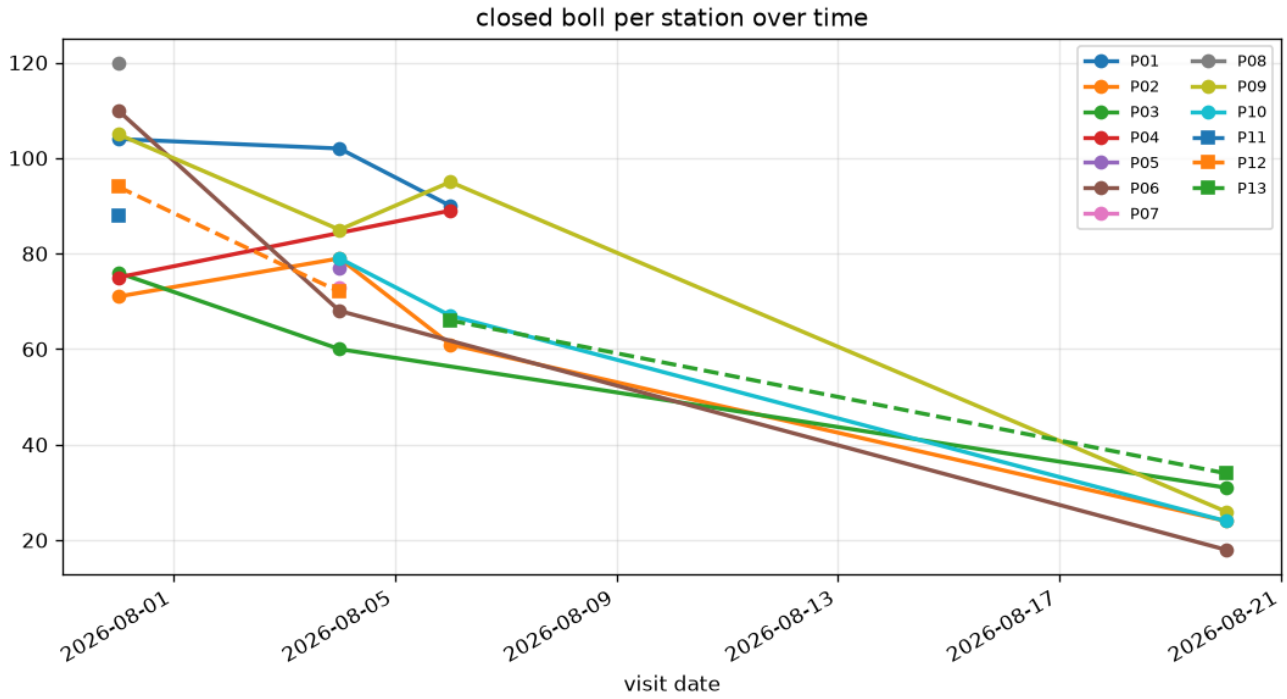
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\station_check\stations_map.png

Open bolls per plant over time (hand labels + Roboflow boll model)



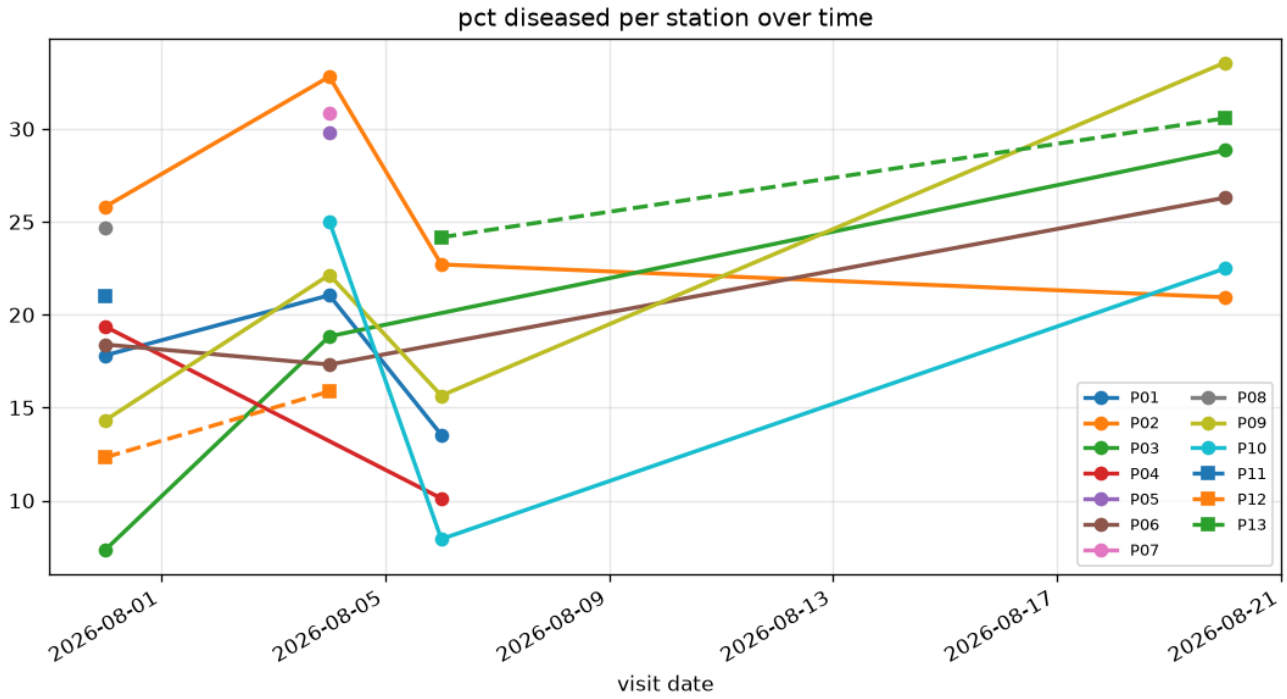
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\open_boll.png

Closed bolls per plant over time



File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\closed_boll.png

% diseased leaves per plant over time



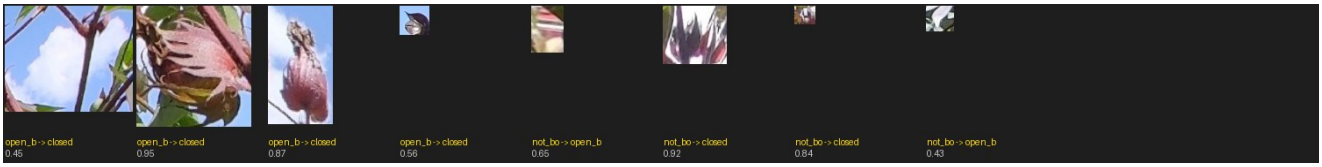
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\pct_diseased.png

Hand labels moved from the panorama onto the 4-POV views (check sheet)



File: F:\Project and completed courses\Agri EYE 360\agrieye_output\pov_dataset\check_projection.jpg

Roboflow boll model: every mistake on the 355 test images



File: F:\Project and completed courses\Agri EYE 360\sample_results\roboflow_boll_test\mistakes.jpg

Open (cyan, solid) and closed (magenta, dashed) bolls on the 4 views with the most open bolls



File: F:\Project and completed courses\Agri EYE 360\sample_results\boll_figures\00_open_vs_closed.jpg

SAM 3 leaf outlines (P09, 31 Jul, east view): every outline is a real leaf, but some leaves have none



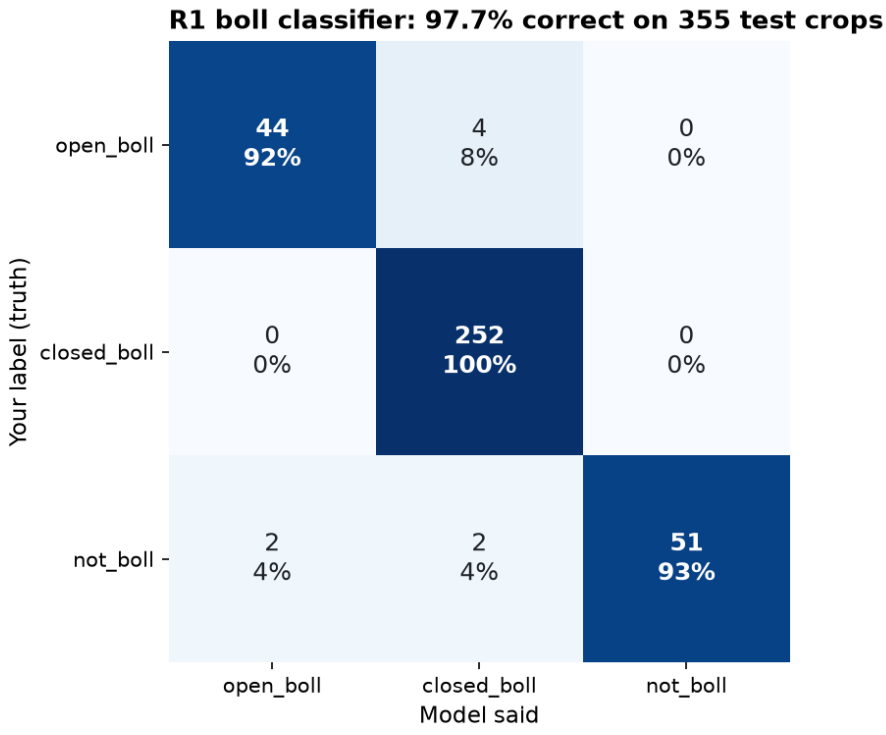
File: F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\example_P09_0731_E.jpg

One-step boll detector on a test view (cyan = open, magenta = closed, white = hand labels)



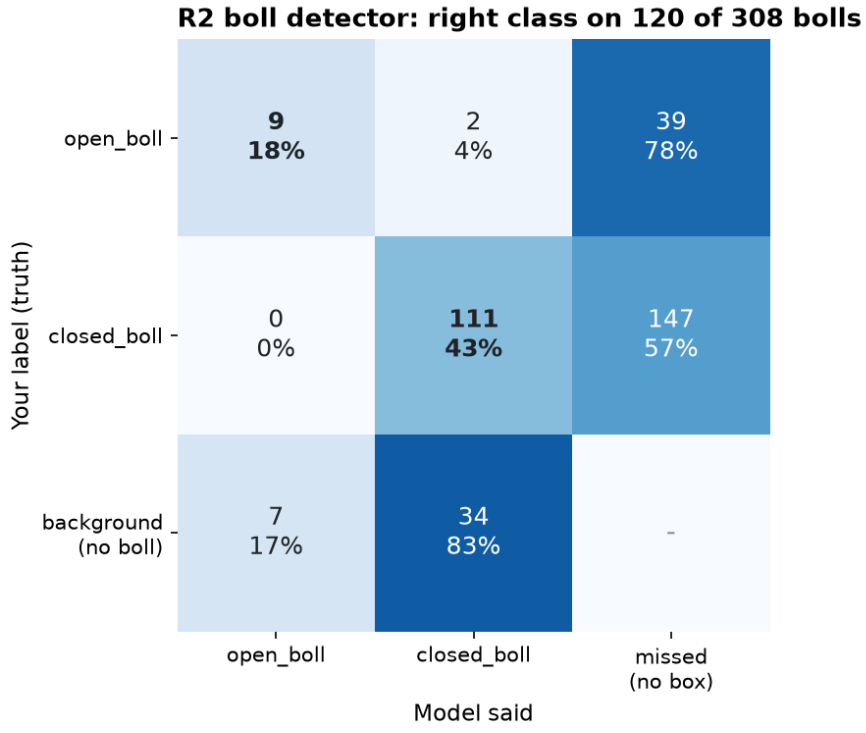
File: F:\Project and completed courses\Agri EYE 360\sample_results\roboflow_boll_detect_test\preview_P13_2026-08-20_20260820_140906_E.jpg

R1 boll classifier: confusion matrix on 355 test crops



File: F:\Project and completed courses\Agri EYE 360\sample_results\confusion_matrices\R1_confusion_matrix.png

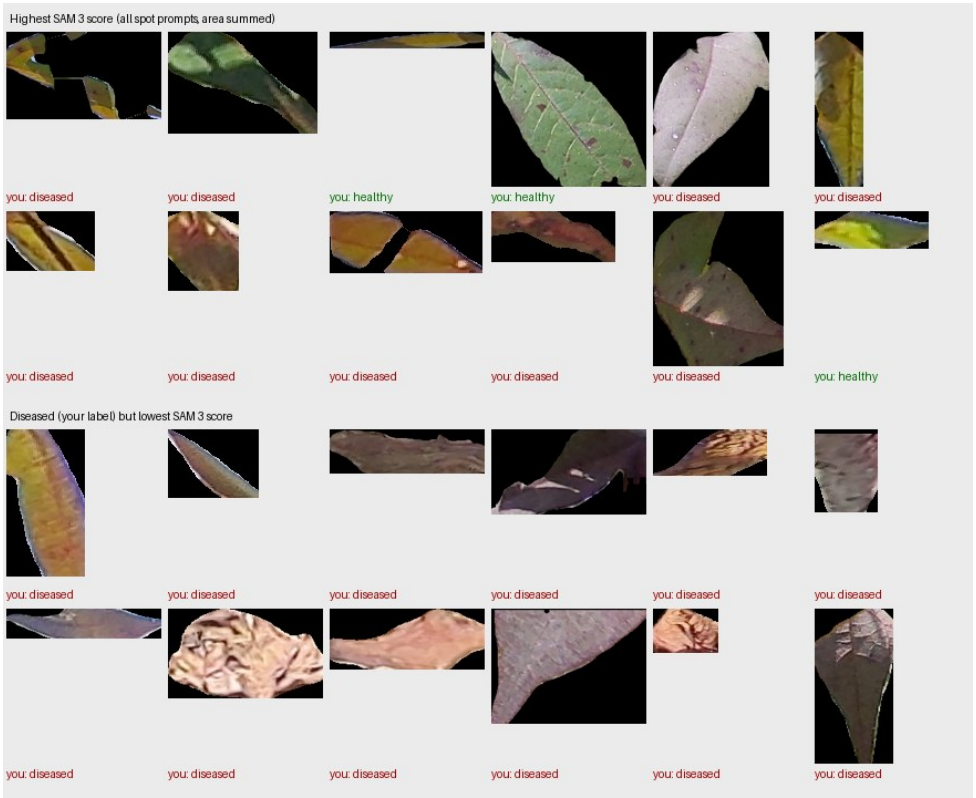
R2 boll detector: confusion matrix on 18 test views



cotton-bolls-field-view-trbz1/1 · 18 test views · model boxes at confidence >= 0.5, IoU >= 0.5

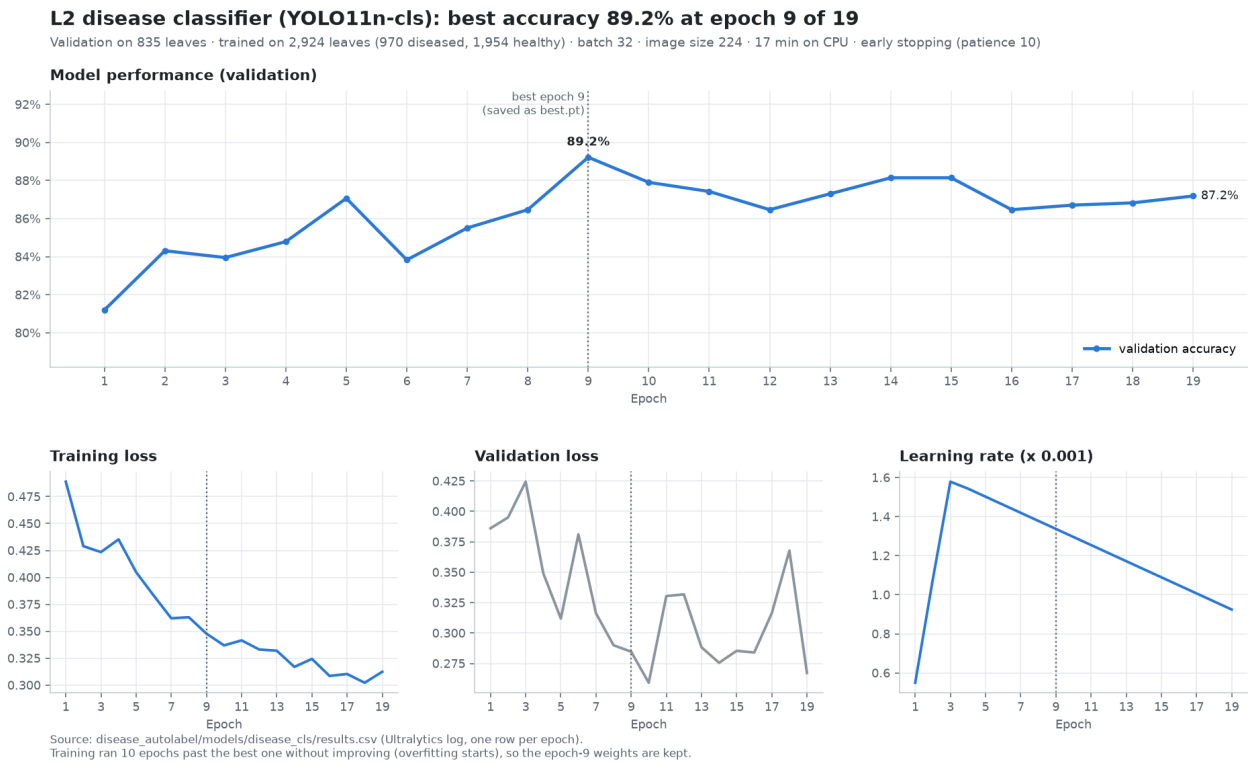
File: F:\Project and completed courses\Agri EYE 360\sample_results\confusion_matrices\R2_confusion_matrix.png

SAM 3 alone, disease-spot prompts: highest-scored leaves (top) and diseased leaves it missed (bottom)



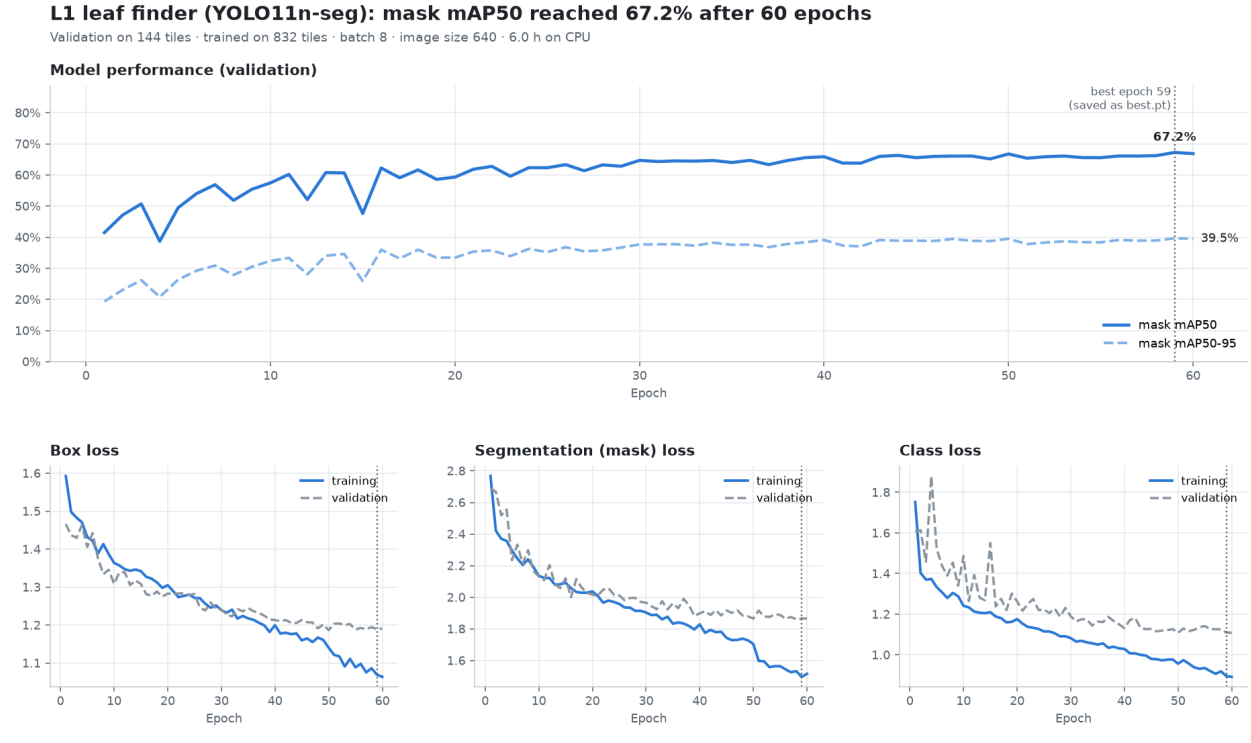
File: F:\Project and completed courses\Agri EYE 360\sample_results\sam3_spot_test\examples.jpg

L2 disease classifier: training per epoch



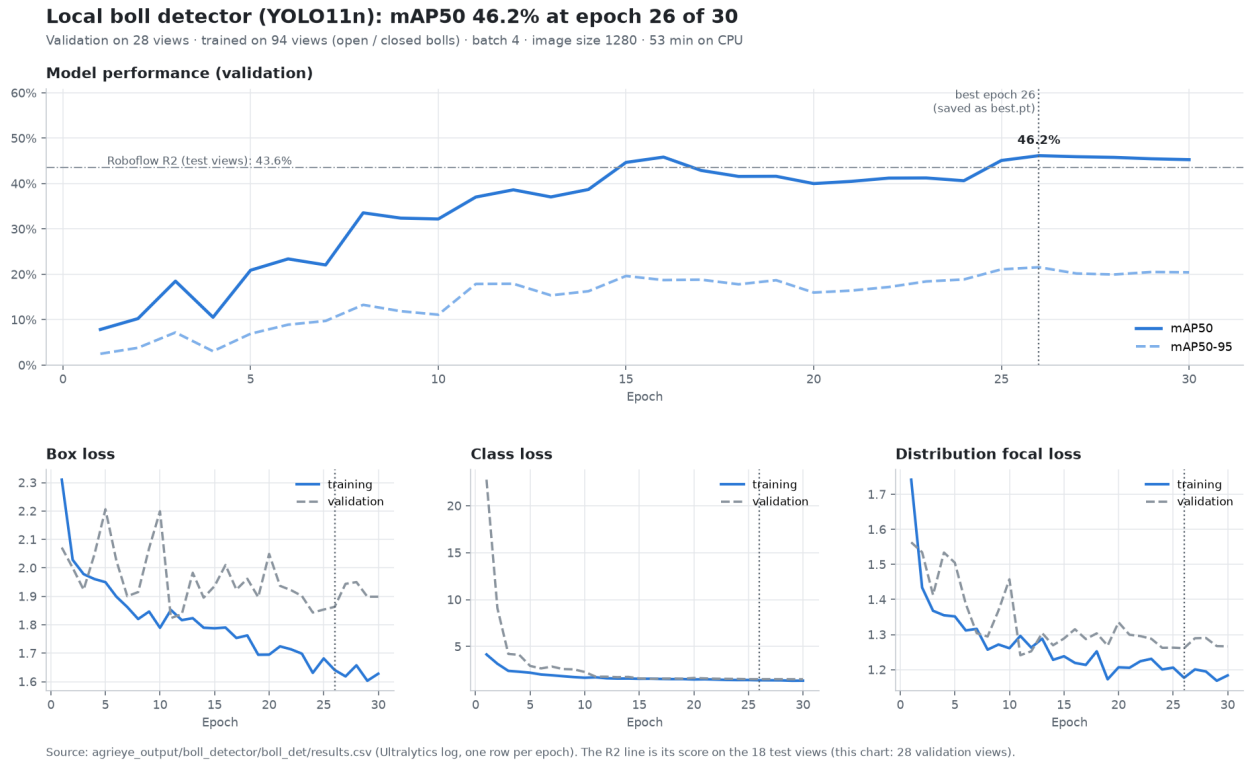
File: F:\Project and completed courses\Agri EYE 360\sample_results\training_graphs\L2_disease_classifier_training.png

L1 leaf finder: training per epoch



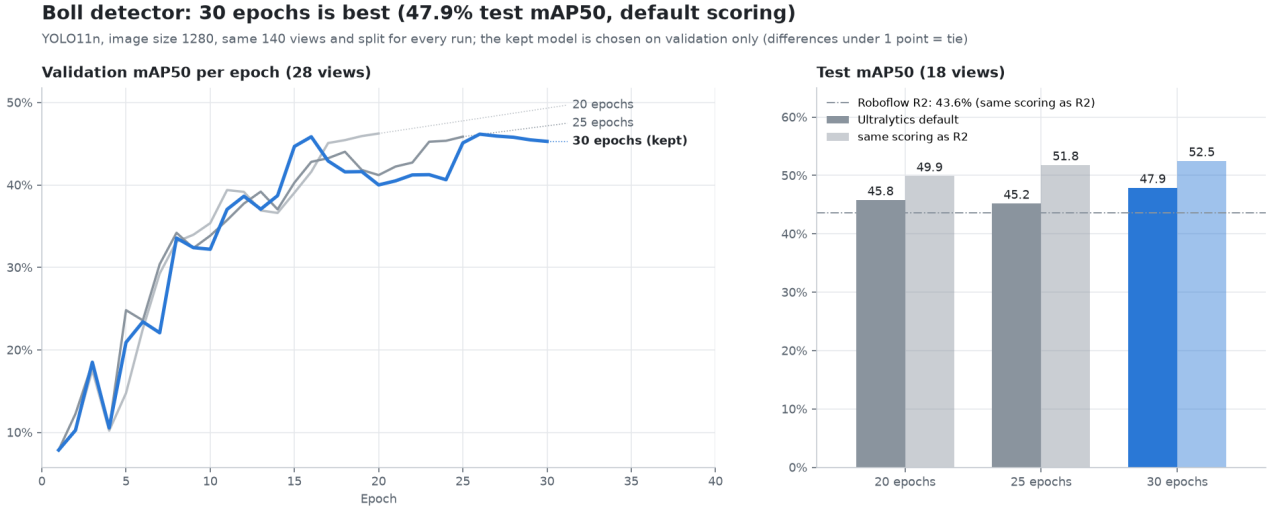
File: F:\Project and completed courses\Agri EYE 360\sample_results\training_graphs\L1_leaf_finder_training.png

Local boll detector (YOLO11n, 30 epochs, 1280 px): training per epoch



File: F:\Project and completed courses\Agri EYE 360\sample_results\training_graphs\boll_detector_training.png

Boll detector: 20 vs 25 vs 30 epochs (validation curves and test scores)



File: F:\Project and completed courses\Agri EYE 360\sample_results\training_graphs\boll_detector_epochs_compare.png